Appendix A.

We built Generalized Linear Mixed Models (GLMMs) to study the influence of weather conditions (Table A.1) on the response variable measured as a comparison of pre-migration dates versus departure dates.

The departure date was considered as the response variable that follows a binomial distribution with parameters p_{iy} and n_{iy}, with p_{iy} the probability of migration and n_{iy} the total number of days recorded, for each individual *i* on the year *y* (Eq. 1). The day that each Ruddy-headed goose starts its migration was set as 1, while the previous fifteen days considered were set as 0. To test the effect of different meteorological variables, we included the effect of covariates listed in Table 1 (Eq 2).

Eq. 1 $\quad$ Departure date ~ Binom (p_{iy}, n_{iy})

Eq. 2 $\quad$ logit (p_{iy}) = Σ c_j x covariate$_j$ + rnd (name, year)

With c_j the corresponding parameter to the j covariate and the sum runs over all the covariates considered in Table A.1. We set the identity of each individual, the year and their interaction as random effects, to include possible differences that we did not measure between individuals and years (second term in Eq.2). We tested different fixed-effect variables (Table A.1) with clear ecological meaning for the species (Austin 2002). All tested variables were normalized to make them dimensionless quantities varying between 0 and 1 (Hooten and Hefley 2019). For all models, we fitted the parameters c_i of Eqs.1 and 2, by maximum likelihood and we used a Binomial error distribution. In addition, multicollinearity of potential predictors can make interpretation of alternative models difficult (Lennon 1999). Thus, we used the Spearman rank correlation coefficient (rs) to detect if two variables were correlated (rs > 0.6) (Table A.2). We took into consideration variables with clear ecological meaning for the species (Austin 2007), but in the case that two important variables were correlated, we tested the effect of both variables in separate models. Based on these, we found high correlation between visibility and humidity (rs= -0.64) and included visibility in the final models (Table A.3).

We also checked the variance inflation factors (VIFs) of the variables to assess the extent of any remaining collinearity (Zuur et al. 2009). To do this, we first fit a GLMM with binomial response and logit link function (i.e., a logistic regression model), containing all explanatory variables, to the presence/absence data and then we calculate the VIFs for each variable from the resulting model (Table A.3). We use the *vif* function in R studio (R Development Core Team 2017) using the *car* package (Fox and Weisberg 2019) to calculate the VIFs (Table A.4), which its value resulted below 10 suggesting that collinearity is not a major issue (Zuur et al. 2009).

For the model selection, we started with a full model which included all non-correlated predictors. We used the p-value to eliminate variables (non-significant variables; p-value > 0.5) one by one. Models were compared using the Akaike Information Criteria (AIC) value, the Bayesian Information Criteria (BIC), and Analysis of Variance (ANOVA) test in order to select the best one. For these analyses, we used the *lme4* package (Bates et al. 2015) with the *glmer* function in R studio. From the three competitive models (ΔAIC < 2) taking into consideration different combination of uncorrelated variables, we chose the most parsimonious model (i.e. less parameters), with the best lowest BIC and with all significant variables taking into account the ANOVA results (Table A.3).

Table A.1. Description of the meteorological variables used in the GLMM models developed for Ruddy-headed goose in the southern Pampas, Argentina.

Weather variable	**Variable description**	**Expected relationship with the conditions at the departure day**
Wind velocity	Wind speed (Km/h)	High wind speed
Wind direction	Four major wind directions (4 categories: North, South, East and West)	Favourable tailwind conditions
Cloud cover	Percentage of clouds (%)	Clear sky, low percentage of clouds
Visibility	Visibility (Km)	Higher visibility
Precipitation	Mean precipitation (mm)	Lower precipitations
Humidity	Daily relative humidity (%)	Lower humidity
Air pressure	Atmospheric pressure (Hpa)	Lower pressure
Moonlight	Percentage of the Moonlight (%)	Full moon, high percentage of moonlight.
Temperature	Ground temperature (°C)	Higher temperatures

Table A.2. Spearman-correlation matrix for the meteorological variables used in the GLMM models developed for Ruddy-headed geese in the southern Pampas, Argentina. Two variables were considered correlated when the Spearman rank (rs) > 0.6.

	Temperature	Humidity	Moonlight	Wind velocity	Air pressure	Precipitation	Visibility	Cloud cover
Temperature	1							
Humidity	0.04	1						
Moonlight	0.02	-0.29	1					
Wind velocity	0.07	-0.36	0.06	1				
Air pressure	-0.34	0.17	-0.2	-0.31	1			
Precipitation	0.2	0.29	0.03	0.06	-0.11	1		
Visibility	-0.17	**-0.64**	0.17	0.29	-0.07	-0.18	1	
Cloud cover	0.22	0.51	-0.07	-0.08	-0.2	0.29	-0.45	1

Table A.3. Generalized Linear Mixed Models (GLMMs) obtained by testing different combination of uncorrelated meteorological variables on the response variable, measured as a comparison of pre-migration dates versus departure dates of Ruddy-headed goose geese in the southern Pampas, Argentina. For each competing models, the Akaike Information Criterion (AIC), the difference between AIC of the current model and the most-parsimonious model (ΔAIC), Bayesian Information Criteria (BIC), Explained deviance, Chi-square and p are given.

Models	AIC	ΔAIC	BIC	Deviance	Chisq	Df	Pr(>Chisq)
Wind velocity + Visibility + Cloud cover + (1 | Year)	**131.33**	**0.43**	**151.36**	**121.33**	**5.0787**	**1**	**0.02422**
Wind velocity + Precipitation + Visibility + Cloud cover + (1 | Year)	130.9	0	154.93	118.9	2.4364	1	0.1185
Wind velocity +Air pressure + Precipitation + Visibility + Cloud cover + (1 | Year)	130.97	0.07	159.01	116.97	1.9279	1	0.165
Wind velocity +Air pressure + Precipitation + Temperature + Visibility + Cloud cover + (1 | Year)	133.41	2.51	164.46	116.41	0.563	1	0.4531
Wind Direction + Wind velocity +Air pressure + Precipitation + Temperature + Visibility + Cloud cover + (1 | Year)	134.37	3.47	170.42	116.37	0.0381	1	0.8453
Wind Direction + Wind velocity +Air	136.12	5.22	176.18	116.12	0.2495	1	0.6174

pressure + Precipitation + Temperature + Visibility + Cloud cover +Moonlight + (1 | Year)

Table A.4. Variance inflation factors (VIFs) of the meteorological variables used in the GLMM models developed for Ruddy-headed geese in the southern Pampas, Argentina. VIFs values have to be below 10, suggesting that collinearity is no longer a major issue.

Variables	VIFs
Temperature	1.89
Moonlight	1.11
Wind velocity	1.36
Air pressure	1.47
Precipitation	1.59
Visibility	1.79
Cloud cover	1.72

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